AlZEIHMERS

WHAT IS ALZHEIMERS DISEASE?

Alzheimers Diseases is the most common form of dementia and is categorised by loss of memory and cognitive functions.

One of the major risk factors is age, with elder citizens having symptoms in their mid sixties.

SYMPTOMS

. Memory problems

. Decline in Visual-spatial skills

. Behavioural Changes

. Easily lose personal objects

. Difficulty with tasks



WAYS THE BRAIN CHANGES

. Decreased connection or synapses.

. Reduction in glucose and energy production for metabolic activity.

. Decreased neurogenesis in areas for memory.

Although the brain normally shrinks in ageing, for Alzheimer patients, they lose neuronal connections, metabolic activity and neurogenensis functions as it gets worse, which can make it fatal.



Main Characteristics

. Amyloid Plaques - clumps in the brain by Beta-Amyloid protein 42.

. Tau Tangles - Normal tau molecules detach from microtubules and attaches abnormally to other molecules, forming tangles.

. Loss of neural functions and synapses leads to brain atrophy or cell death.

. Abnormal functioning or defunctioning of Microglia or Astrocytes; cells that protect neurons, leads to failure in debris cleanup, causing buildups.



THANK YOU!

1) https://www.nia.nih.gov/health/alzheimers-and-dementia/what-alzheimers-disease

2) https://www.nia.nih.gov/health/alzheimers-causes-and-risk-factors/what-happens-brain-alzheimers-disease